Benchmarking and Adaptive Optimization of Filtered Vector Search on Proximity Graphs

Preliminary Research Proposal

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1. RESEARCH MOTIVATION AND PROBLEM

Graph-based indexes such as HNSW have become the dominant approach for approximate nearest neighbor search (ANNS) in vector database systems. However, real-world vector retrieval workloads increasingly require *filtered* queries—retrieving the top-*k* vectors most similar to a query while simultaneously satisfying structured metadata predicates (e.g., `brand = "Nike" AND price < 50`). This filtered ANNS problem is central to production vector databases powering retrieval-augmented generation (RAG), recommendation, and hybrid search applications.

The core challenge is that metadata filters interact with graph traversal in complex ways. Under highly selective filters, large portions of the proximity graph become ineligible, disrupting greedy search connectivity and degrading both recall and latency. Recent systems—including Filtered-DiskANN, ACORN, NaviX, JAG, and others—each propose different structural or algorithmic solutions. I am particularly interested in characterizing how these filtered-ANNS methods behave under different selectivity, data-distribution, and CPU execution conditions, and in identifying performance regimes that may benefit from further optimization.

2. RESEARCH QUESTIONS

- How do representative filtered ANNS methods compare in recall–latency trade-offs as filter selectivity varies from permissive to highly restrictive?
- Under what conditions do graph-based filtered search methods exhibit disproportionate traversal overhead or recall degradation?
- Can query-time adaptive strategies reduce unnecessary search work (e.g., redundant distance computations or excessive traversal) while maintaining comparable retrieval quality?

3. PROPOSED RESEARCH DIRECTION

My current thesis direction is **filtered vector search on proximity graphs**, focused on understanding and improving how graph-based ANNS indexes handle metadata-constrained queries. The research landscape has developed rapidly: HNSW (Malkov & Yashunin, 2020) established the foundation for graph-based ANNS; Filtered-DiskANN (Gollapudi et al., 2023) introduced filter-aware graph construction; ACORN (Patel et al., 2024) proposed predicate-agnostic subgraph construction for hybrid queries; NaviX (Sehgal & Salihoglu, 2025) studies adaptive search under varying selectivity and query correlation; and JAG (Xu et al., 2026) targets robust performance across the full selectivity spectrum through joint attribute-vector graph construction.

This direction connects directly to active research in vector database system design, query optimization and execution, adaptive indexing, and systems infrastructure for modern AI workloads—areas where rigorous empirical understanding of filtered search behavior can inform both index design and query-processing strategies.

4. METHODOLOGY

The research follows a diagnostic-driven, three-phase approach:

Phase 1 — Baseline Reproduction. Study and reproduce representative filtered vector search methods (beginning with HNSW with pre/post-filtering, progressing to available open-source implementations such as Filtered-DiskANN and ACORN). Establish reliable baselines on standard benchmark datasets with controlled filter predicates.

Phase 2 — Empirical Diagnosis. Systematically benchmark baselines across a selectivity spectrum—from permissive filters (retaining >80% of candidates) to highly restrictive filters (<1%). Measure recall@k, query latency, distance computations per query, and candidate set sizes. Identify reproducible performance regimes where traversal overhead scales disproportionately or recall degrades sharply.

Phase 3 — Targeted Optimization. Based on evidence from Phase 2, design and evaluate a lightweight query-time adaptation mechanism targeting one specific identified bottleneck. The precise optimization strategy will be defined after baseline reproduction and empirical diagnosis are complete.

Note: *The final algorithmic contribution will be refined through literature study, baseline implementation, and experimental evidence. The current working hypothesis—that query-time adaptation can reduce redundant traversal work under selective filters—serves as a research direction, not a proven claim.*

5. EXPECTED CONTRIBUTION

- A reproducible empirical comparison of filtered ANNS methods on proximity graphs under controlled selectivity conditions, with CPU-oriented performance analysis.
- A diagnostic characterization of performance bottlenecks as a function of filter selectivity and data distribution.
- A candidate adaptive optimization strategy, evaluated against established baselines, with honest reporting of both improvements and limitations.

The scope is calibrated for an undergraduate research project. The primary value lies in rigorous benchmarking and diagnosis; the optimization component will be pursued to the extent supported by experimental evidence.

6. CURRENT PREPARATION

- Database systems fundamentals through CMU 15-445/645, covering storage architecture, buffer management, page organization, and memory hierarchy.
- Survey of the filtered ANNS research landscape, including HNSW, Filtered-DiskANN, ACORN, NaviX, JAG, and recent benchmarking work.
- Strong algorithmic foundation: 2600+ solved DSA problems and systems-level project experience (embedded C++ firmware, database-backed applications), providing the implementation discipline required for baseline reproduction and experimental work.

Relevant coursework includes Data Structures & Algorithms, Database Systems, Operating Systems, Networking, and OOP. Primary implementation languages: C++ and Python. I have not

yet implemented HNSW or advanced filtered-ANN systems; Phase 1 is designed to build this capability systematically under guidance.

REFERENCES

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This proposal outlines a current research direction and working plan. The specific algorithmic contribution and experimental scope will be refined through further literature study, baseline implementation, and empirical analysis, ideally under the guidance of a research mentor.